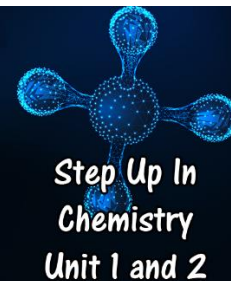


4.2 Stoichiometry

Problems Worksheet



1. Hydrogen sulfide reacts with zinc oxide to produce water and zinc sulfide.
 - a. Write an equation for the reaction.

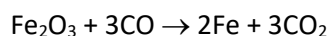
In an experiment, 8.90 g of zinc oxide reacts.

- b. Determine the number of moles zinc sulfide produced.

- c. Determine the mass of hydrogen sulfide required.

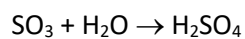
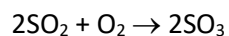
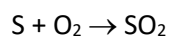
- d. Determine the mass of water produced.

2. A precipitate of lead(II) iodide is produced when lead(II) nitrate reacts with sodium iodide.
- Write an equation for the reaction.
 - Determine the mass of lead(II) iodide produced if 9.70 g of lead(II) nitrate reacts with excess sodium iodide.
3. Iron ore is comprised mostly of the mineral haematite, which is iron(III) oxide (Fe_2O_3). The iron is extracted from the ore by a reaction with carbon monoxide in a blast furnace. The overall equation for the process is shown:



- In WA, approximately 2.26 tonnes of haematite is mined each day. Calculate the mass of iron which could be produced from this ore.
- What mass of carbon monoxide would be required to extract the iron?
- What mass of carbon dioxide would be produced?

4. Sulfuric acid is made by a series of chemical reactions in the Contact Process. The equations are:

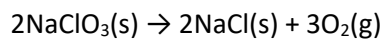


- a. The engineers at a sulfuric acid plant want to produce 1.00 tonnes of sulfuric acid per day. Calculate the mass of sulfur trioxide they will need to produce each day.

- b. What mass of sulfur will they need?

- c. Extension: combine the three equations into one overall equation for the process.

5. In a submarine, aircraft or spacecraft, oxygen can be generated in an emergency with an 'oxygen candle' containing sodium perchlorate:



- a. Determine the mass of oxygen gas produced by 1.000 kg of sodium perchlorate.

- b. What mass of sodium chloride will be produced in this reaction?

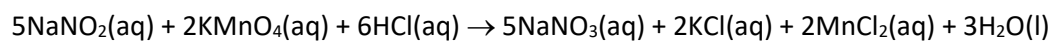
An alternative oxygen candle uses the following reactin:



- c. Determine whether this candle would be more or less effective at producing oxygen than the sodium perchlorate candle. Use a calculation to justify your answer.

6. Sodium chloride can be produced in an explosive reaction between sodium metal and chloride gas. In an experiment, 43.0 g of sodium is reacted with 75.1 g of chlorine gas.
- Determine the limiting reagent.
 - Determine the mass of sodium chloride produced.
 - What mass of the excess reagent will be left over after the reaction?

7. Sodium nitrite can react with potassium permanganate and hydrochloric acid according to the following equation:



Determine the mass of manganese chloride produced if 5.00 g of NaNO_2 reacts with 8.50 g of HCl .